22/01/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 6

Topics covered :

Physics : Electric Charges & Field, Electrostatic Potential and Capacitance
Chemistry : Solid state, Solutions and Electrochemistry
Botany : Plant Growth and Development, Reproduction in Organisms
Zoology : Chemical Coordination and Integration, Reproduction in Organisms

Instructions :

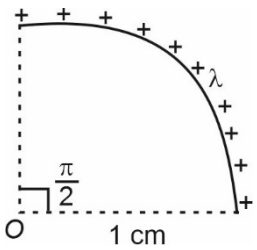
(i) Use blue/black ballpoint pen only to darken the appropriate circle.
(ii) Mark should be dark and should completely fill the circle.
(iii) Dark only one circle for each entry.
(iv) Dark the circle in the space provided only.
(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.
(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. Two negative point charges of unequal magnitude are placed at two points $A(0, 0, -2)$ and $B(0, 0, 2)$. A small negative charge q is placed at null point, then select correct statement.
   (1) The charge q is not in equilibrium
   (2) The charge q is in stable equilibrium along z-axis
   (3) The charge q is in stable equilibrium along x-axis
   (4) Both (1) and (3) are correct

2. The electric field at point O of a circular arc of radius 1 cm having linear charge density 2 C/cm is given as



   (1) 18×10^{14} V/m
   (2) 1.8×10^{14} V/m
   (3) 2.5×10^{14} V/m
   (4) 12×10^{14} V/m

3. Two pith balls each of mass m carrying equal charges are suspended from a common point by strings of equal length. At the equilibrium, separation between the balls is r. If the angular separation between the two strings is $\frac{\pi}{3}$, then the charge on each ball is
   (1) $\left[4\sqrt{3}\pi\varepsilon_0 mgr^2\right]^{1/2}$
   (2) $\left[\frac{4\pi\varepsilon_0 mgr^2}{\sqrt{3}}\right]^{1/2}$
   (3) $\left[\frac{\sqrt{3}mgr^2}{4\pi\varepsilon_0}\right]^{1/2}$
   (4) $\left[\frac{mgr^2}{4\sqrt{3}\pi\varepsilon_0}\right]^{1/2}$

4. If two identical spheres having charges $6\mu C$ and $-2\mu C$ are separated by certain distance then the force is F. A third sphere of the same size but uncharged is brought in contact with A, then brought in contact with the second, and finally removed from both. The magnitude of the new force between A and B is

(1) $\frac{3F}{8}$ (2) F

(3) $\frac{F}{4}$ (4) $\frac{F}{8}$

5. The electric field in a region is given as $\vec{E} = \left(3\hat{i} + 4\hat{j} + \hat{k}\right)$ N/C. The flux of this field linked with a square of surface area 10 cm^2 kept parallel to x-z plane is

(1) 40 N m^2/C (2) Zero

(3) 30 N m^2/C (4) 4×10^{-3} N m^2/C

6. A point charge of 2 C is placed at a distance 10 cm directly above the centre of a square of side 20 cm. The flux through the square is

(1) $\frac{1}{6\varepsilon_0}$ (2) $\frac{1}{3\varepsilon_0}$

(3) $\frac{1}{\varepsilon_0}$ (4) Zero

7. Two large thin metal plates are kept parallel and close to each other. On their inner faces, the plates have surface charge densities of opposite signs and of same magnitude. If the electric field between the plates is 1.2×10^{-10} N/C. Then the magnitude of the surface charge density is nearly

(1) 1.1×10^{-21} C/m^2

(2) 6×10^{-23} C/m^2

(3) 3.2×10^{-23} C/m^2

(4) 4×10^{-23} C/m^2

8. Three charges q_1 = 1 μC, q_2 = 4μC and q_3 = –5 μC are placed at points $A(0, \sqrt{3}, 0)$ mm, $B(-1, 0, 0)$ mm and $C(1, 0, 0)$ mm respectively. The magnitude of the dipole moment of the system is

(1) 8×10^{-9} C m

(2) 4.6×10^{-9} C m

(3) 2.3×10^{-9} C m

(4) 9.2×10^{-9} C m

9. There is a uniform electric field directed along positive z-axis. If an electron moves in the negative z-axis then its electric potential energy

(1) Decreases

(2) Increases

(3) Remains constant

(4) Depends on velocity

10. An alpha particle and proton are placed at rest in a uniform electric field E and then released. The ratio of kinetic energy attained by the alpha particle and proton after moving equal distance, is

(1) $\frac{1}{2}$ (2) 4

(3) $\frac{1}{4}$ (4) 2

11. A non-conducting solid sphere of radius R is uniformly charged. The magnitude of the electric field due to the sphere at a distance r from its centre is

(1) Directly proportional to r^2 for $r < R$

(2) Inversely proportional to r^2 for $0 < r < \infty$

(3) Is continuous at $r = R$

(4) Is discontinuous at $r = R$

12. Two short dipole of moments $\vec{p_1} = p\hat{i}$ and $\vec{p_2} = p\hat{j}$ are placed at origin. The magnitude of electric field at a point (1, 0, 0) m is $[p = \sqrt{5} \times 10^{-8}$ C m]

(1) 202 V/m (2) 404 V/m

(3) 450 V/m (4) 150 V/m

13. A hollow metallic sphere of radius 20 cm is charged such that the potential of its surface is 50 V. The potential at a distance 10 cm from the centre of the sphere is

(1) 25 V (2) 50 V

(3) Zero (4) 1000 V

14. An oil drop of 16 excess electrons is held stationary under a constant electric field E. If the density of the oil is ρ and charge of electron is e, then the radius of the drop is

(1) $\left[\frac{12eE}{\pi\rho g}\right]^{1/3}$ (2) $\left[\frac{12eE}{\pi\rho g}\right]^{1/2}$

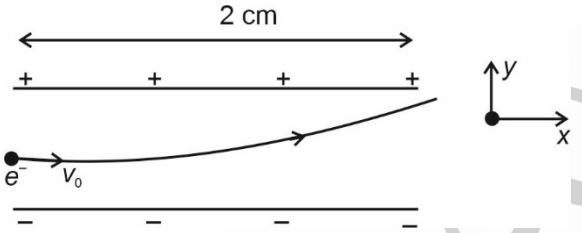
(3) $\left[\frac{6eE}{\pi\rho g}\right]^{1/3}$ (4) $\left[\frac{48eE}{\pi\rho g}\right]^{1/2}$

15. A light rod of length L with two point charges q and $-q$, both of mass m are attached at its end is placed in a region of uniform electric field with intensity E. The rod can rotate in the horizontal plane about a fixed axis passing through its mid point. If it is slightly displaced from unstable equilibrium then the maximum angular velocity of the rod is

(1) $\sqrt{\frac{qE}{mL}}$ (2) $\sqrt{\frac{2qE}{mL}}$

(3) $2\sqrt{\frac{2qE}{mL}}$ (4) $\sqrt{\frac{4qE}{mL}}$

16. The distance between the two plates of a parallel plate capacitor is 1 cm and potential difference between them is 10^3 volts. If an electron of energy 1500 eV enters at right angles to the uniform electric field then the deflection of the electron along y-axis is [The length of the plate = 2 cm]



(1) 3.3 mm (2) 6.7 mm

(3) 1.4 mm (4) 2.3 mm

17. The conducting spheres of radii 4 cm and 8 cm are given the same charge of 10 nC. If two spheres are brought in contact and separated then the potential of the larger sphere will be

(1) 3×10^3 V (2) 1.5×10^3 V

(3) 10^3 V (4) 7.5×10^3 V

18. Two capacitors of capacitance 4 μF and 8 μF are charged to a potential of 16 V each. If they are connected to each other with positive plate of one joined to negative plate of other then the percentage of loss of energy will be nearly

(1) 50% (2) 75%

(3) 80% (4) 89%

19. A parallel plate air capacitor is charged by a battery. After disconnecting the charging battery the distance between the plates of the capacitor is doubled (Using an insulating handle)

The electric field between the plates will be

(1) Halved (2) Doubled

(3) Constant (4) Quadrupled

20. Two capacitors each of 4 μF capacitance are connected in parallel, then they are charged by a battery of 100 V. If space between both capacitors is filled with a dielectric medium of constant (K = 4) then the total potential energy stored in the capacitors will be

(1) 0.16 J (2) 0.08 J

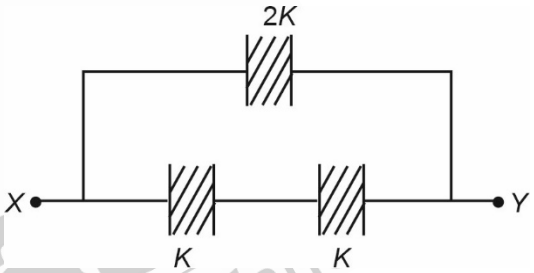
(3) 1.2 J (4) 2.4 J

21. A cylindrical capacitor has two coaxial cylinders of length 10 cm and radii 1 cm and 2 cm. The outer cylinder is earthed and the inner cylinder is given a charge of 2 μC. The potential of the inner cylinder is nearly

(1) 3×10^5 V (2) 2×10^5 V

(3) 3.2×10^5 V (4) 2.5×10^5 V

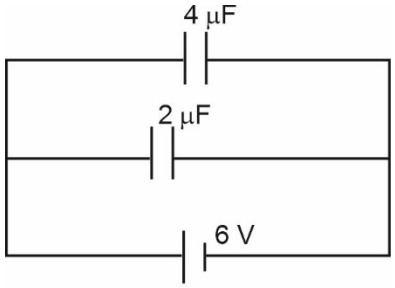
22. Three parallel plate capacitor with each of capacitance C are filled with dielectric constant K, K and $2K$ as shown in figure. If equivalent capacitance between points X and Y is $4C$ then the value of K is



(1) $\frac{1}{5}$ (2) $\frac{8}{5}$

(3) $\frac{1}{4}$ (4) $\frac{3}{44}$

23. Two uncharged capacitors of capacitance 2μF and 4μF connected by a battery as given in the figure. The work done by the battery in charging will be



(1) 216 J (2) 108 J

(3) 108 μJ (4) 216 μJ

24. Two charges $2\mu C$ and $-2\mu C$ are placed at (10 cm, 0, 0) and (20 cm, 0, 0) respectively in the presence of a uniform external electric field of 2×10^4 V/m directed towards positive *x*-axis. The total electrostatic energy of the configuration will be

(1) –356 J (2) –360 J

(3) –360 mJ (4) –356 mJ

25. Two charged conducting spheres of radii 10 cm and 25 cm are connected to each other by a conducting wire. The ratio of electric field at the surfaces of the two spheres will be

(1) 2 (2) $\frac{5}{2}$

(3) $\frac{3}{2}$ (4) $\frac{25}{4}$

26. Two parallel plates have area *A* and separation *r*. They are given charges *q* and $\frac{q}{2}$. The force acting between the plates is

(1) $\frac{q^2}{4A\varepsilon_0}$ (2) $\frac{q^2}{A\varepsilon_0}$

(3) $\frac{q^2}{2A\varepsilon_0}$ (4) $\frac{2q^2}{A\varepsilon_0}$

27. A spherical capacitor has an inner sphere of radius *R* and an outer sphere of radius $\frac{3R}{2}$. The outer sphere is earthed and the inner sphere is given a charge *q*. If the space between the concentric spheres is filled with a liquid of dielectric constant $K = 2$, then the potential of the inner sphere is

(1) $\frac{q}{4\pi\varepsilon_0 R}$ (2) $\frac{q}{\pi\varepsilon_0 R}$

(3) $\frac{q}{24\pi\varepsilon_0 R}$ (4) $\frac{q}{12\pi\varepsilon_0 R}$

28. Some charge is being given to a non-conducting sphere and it is uniformly distributed in its volume, then its potential is

(1) Maximum at surface

(2) Maximum at centre

(3) Remains same throughout the sphere

(4) Maximum somewhere between surface and centre

29. In bringing a proton towards another proton, the electrostatic potential energy of the system

(1) Becomes zero

(2) Increases

(3) Decreases

(4) Remains same

30. The electric potential at a point (*x, y, z*) is given by $V = -xy$ volt. The electrostatic force acting on 1C charge at a point (1, 2, –4) m will be

(1) $\left(\hat{i} + 2\,\hat{j}\right)$ N

(2) $\left(2\hat{i} + \hat{j}\right)$ N

(3) $\left(\hat{i} + 2\,\hat{j} - 4\hat{k}\right)$ N

(4) $\left(\hat{i} - 2\,\hat{j} + 4\hat{k}\right)$ N

31. A charge of 8C is located at the origin. The work done in taking another charge of –2C from a point *A*(0, 0, –4) m to *B*(0, 4, 0) m, is

(1) 72×10^9 J

(2) 36×10^9 J

(3) Zero

(4) -36×10^9 J

32. An electron is released from rest in a uniform electric field. The electric potential energy of the electron

(1) Increases because the electron moves along the electric field

(2) Decreases because the electron moves opposite to the electric field

(3) Remains constant due to uniform electric field

(4) Increases because the electron moves opposite to the electric field

33. A parallel plate capacitor has a uniform electric field in the space between the plates. The electric field due to one plate is *E*. If the distance between the plates is *d* and area of each plate is *A*. Then the energy density stored between the plates of capacitor will be

(1) $\varepsilon_0 E^2 Ad$ (2) $2\varepsilon_0 E^2$

(3) $\frac{\varepsilon_0 E^2}{2}$ (4) $4\varepsilon_0 E^2$

34. An electrical technician requires a capacitance of 4 μF in a circuit across a potential difference of 1000 V. A large number of 2 μF capacitors are available to him, each of which can withstand a potential difference of not more than 200 V. The minimum number of capacitors required are

(1) 50

(2) 100

(3) 64

(4) 18

35. A bullet of mass 10 g having a charge of 1 mC is at rest. The accelerated voltage required to acquire a speed of 5 m/s, is

(1) 12.5 V

(2) 120 V

(3) 125 V

(4) 1.25 V

36. The charge is distributed uniformly in the volume of a sphere of radius 20 cm such that the potential of its surface is 50 V. The potential at the centre of the sphere will be

(1) 75 V

(2) 25 V

(3) 50 V

(4) Zero

37. A charge of 2 μC is given to a conducting sphere of radius 10 cm. The total energy stored in it is

(1) 0.18 mJ

(2) 18 J

(3) 1.8 J

(4) 0.18 J

38. A positive point charge is placed outside the neutral conducting sphere of radius R at a distance $2R$ from the centre. If potential is zero at infinite then the net potential of the sphere is

(1) Positive (2) Negative

(3) Zero (4) Infinite

39. An electric field is given as $\vec{E} = (\hat{i} + 2\hat{j})$ V/m . The potential difference (V_A – V_B) between points A(1, 1, 0) m and B(2, 2, 0) m is

(1) –3 V (2) –6 V

(3) 3 V (4) Zero

40. A parallel plate capacitor of capacitance 40 μF is connected to a battery of emf 5 V. If a dielectric material of dielectric constant K = 4 is inserted between the plates, the magnitude of the induced charge will be

(1) 200 μC (2) 600 μC

(3) 1200 μC (4) 400 μC

41. A conducting sphere having a charge 2 μC is surrounded by an uncharged concentric conducting spherical shell. The potential difference between the surface of solid sphere and the shell is 5 V. If the shell is given a charge –4 μC, then the new potential difference between the same surfaces will be

(1) 5 V (2) 10 V

(3) 20 V (4) –10 V

42. A short electric dipole of moment $\vec{p} = p\hat{i}$ is placed at origin. The electrostatic potential at a point (–a, 0, 0) is

(1) $\frac{-p}{4\pi\varepsilon_0 a}$ (2) $\frac{p}{4\pi\varepsilon_0 a^2}$

(3) $\frac{-p}{4\pi\varepsilon_0 a^2}$ (4) $\frac{2p}{4\pi\varepsilon_0 a^2}$

43. The equipotential surface due to the radial electric field will be

(1) Spherical (2) Cylindrical

(3) Planar (4) Circular

44. A small positive test charge is released at rest at a point in an uniform electric field. It will travel

(1) Along the field line

(2) Opposite to the field line

(3) Perpendicular to the field line

(4) Along the parabolic path

45. A proton and alpha particle are accelerated from rest through a potential difference V in vacuum. The ratio of final velocity will be

(1) $\sqrt{2}$

(2) 2

(3) $2\sqrt{2}$

(4) 4

CHEMISTRY

46. Which of the following concentration terms is independent of temperature?
(1) Molarity (2) Normality
(3) % w/w (4) % w/V

47. Molality of an aqueous solution of urea in which mole fraction of urea is 0.2
(1) 3.89 m (2) 13.89 m
(3) 5.33 m (4) 18.33 m

48. If O_2 (g) is bubbled through water at 293 K. Calculate the millimoles of O_2(g) which would dissolve in 10 mol of water at oxygen gas pressure of 0.5 atm ($K_H(O_2)$ at 293 K is 3.5×10^4 atm).
(1) 0.143 (2) 0.234
(3) 1.2 (4) 14.3

49. If molality of a dilute solution is doubled then the value of molal elevation constant (K_b) becomes
(1) Doubled (2) Halved
(3) Tripled (4) Unchanged

50. Which thermodynamic property is incorrect for dilute solution of ethyl alcohol and water?
(1) $\Delta H_{mix} > 0$ (2) $\Delta S_{mix} > 0$
(3) $\Delta V_{mix} > 0$ (4) $\Delta G_{mix} > 0$

51. Which of the following aqueous solution will show highest boiling point when each is 0.2 M?
(1) K_2SO_4 (2) NH_2CONH_2
(3) $AlPO_4$ (4) Na_3AlF_3

52. If two volatile liquids $A(P_A^o = 400\ mmHg)$ and $B(P_B^o = 300\ mmHg)$ are mixed in the molar ratio of 2 : 5 then vapour pressure of A in liquid mixture will be
(1) 400 mm Hg (2) 300 mm Hg
(3) 328.57 mm Hg (4) 114.3 mm Hg

53. 0.2 molal aqueous solution of $K_3[Fe(CN)_6]$ ionizes to 40%. Depression in freezing point of solution will be ($K_f(H_2O)$ = 1.86 K molal^{-1})
(1) 0.82 K (2) 0.45 K
(3) 1.23 K (4) 0.63 K

54. If 0.25 molar aqueous solution of Na_2SO_4 is isotonic with 7.5% $\left(\frac{w}{v}\right)$ solution of non-electrolyte and non-volatile solute at 298 K then molar mass (in g mol^{-1}) of solute is
(1) 10
(2) 100
(3) 90
(4) 400

55. van't Hoff factor for Na_2SO_4 solution having 60% dissociation is
(1) 3
(2) 2.2
(3) 1.8
(4) 2.5

56. Among the following which mixture form solution of high boiling azeotrope?
(1) Benzene and toluene
(2) n-hexane and n-heptane
(3) Hydrochloric acid and water
(4) Acetone and Carbon disulphide

57. For solution of concentration of 0.01 m, which one has highest value of van't Hoff factor?
(1) NaCl
(2) KCl
(3) $MgSO_4$
(4) K_2SO_4

58. Observed molar mass of acetic acid which is dimerizes 75% in benzene solvent, is
(1) 60 (2) 96
(3) 37.5 (4) 30

59. In which of the following boiling point of solution will not be increased?
(1) Addition of NaCl in aqueous solution of urea
(2) Addition of Na_2SO_4 in aqueous solution of glucose
(3) Addition of HgI_2 in aqueous solution of KI
(4) Addition of KNO_3 in aqueous solution of sugar

60. Select the correct statement among the following.
(1) Plant cell when placed in hypotonic solution, it will shrink
(2) Osmotic pressure associated with 0.9% mass/volume of NaCl solution is equivalent to fluid inside the blood cell
(3) As compared to other colligative properties, magnitude of elevation in boiling point is large even for very dilute solution
(4) Molal elevation constant always larger than molal depression constant

61. Reduction potential of hydrogen electrode which is placed in solution of pH = 1 is
(1) 1 V (2) – 0.0591 V
(3) – 0.591 V (4) 0.0591 V

62. For a cell involving two electrons, $E^{o}_{cell} = 0.59\,V$ at 298 K, the equilibrium constant for the cell reaction is
(1) 10^5 (2) 10^8
(3) 10^{20} (4) 10^{10}

63. During the electrolysis of an aqueous solution of $AgNO_3$, amount of metal deposited when 0.965 A current will pass for 300 s, is (g atomic mass of silver is 108 g mol^{-1})
(1) 0.42 g (2) 0.324 g
(3) 4.2 g (4) 3.24 g

64. On electrolysis of aqueous solution of $CuSO_4$ using copper electrodes, products at cathode and anode respectively are
(1) Cu and $S_2O_8^{2-}$ (2) Cu and Cu^{2+}
(3) Cu and H_2 (4) Cu and O_2

65. If 2 Faradays of electricity is passed through 1000 ml of 1 M $CuSO_4$ solution. Calculate the molarity of $CuSO_4$ solution after electrolysis.
(1) Zero (2) M
(3) M/2 (4) 2 M

66. Following limiting molar conductivities are given as
$\Lambda^{o}_{m}(Na_3PO_4) = x\,S\,cm^2\,mol^{-1}$
$\Lambda^{o}_{m}(HCl) = y\,S\,cm^2\,mol^{-1}$
$\Lambda^{o}_{m}(NaCl) = z\,S\,cm^2\,mol^{-1}$
$\Lambda^{o}_{m}(\text{in } S\,cm^2mol^{-1})$ for H_3PO_4 will be
(1) $\frac{(x+y-z)}{3}$ (2) $\frac{x+3y-3z}{3}$
(3) x + y – 2z (4) x + 3y – 3z

67. The standard electrode potential (E^o) values of Zn^{2+}/Zn, Cu^{+}/Cu, Au^{3+}/Au and Co^{3+}/Co^{2+} respectively are –0.76 V, 0.52 V, 1.40 V and 1.81 V. Strongest reducing metal/metal ion is
(1) Au (2) Zn
(3) Cu (4) Co^{2+}

68. Faradays charge required for the conversion of 0.2 mol dichromate ion into Cr(III) ion, is
(1) 6 F (2) 1.2 F
(3) 2.4 F (4) 0.6 F

69. Molar conductivity (in S cm^2 mol^{-1}) of 0.01 M aqueous solution of electrolyte whose cell constant and resistance of solution respectively are 1.2 cm^{-1} and 250 Ω is
(1) 480 (2) 240
(3) 520 (4) 600

70. If the electrode potential of $A^{2+} + e^- \longrightarrow A^+$ is x_1 V and of $A^{2+} + 2e^- \longrightarrow A$ is x_2 V then electrode potential of $A^+ + e^- \longrightarrow A$ will be
(1) $(x_1 - x_2)V$ (2) $(x_1 + x_2)V$
(3) $(2x_1 - x_2)V$ (4) $(2x_2 - x_1)V$

71 In which of the following hydrogen concentration cell, cell reaction is spontaneous at 298 K?
(1) $Pt|H_2|H^+(pH = 1) \,||\, H^+(pH = 2)|H_2|Pt$
(2) $Pt|H_2|H^+(pH = 2) \,||\, H^+(pOH = 1)|H_2|Pt$
(3) $Pt|H_2|H^+(pH = 3) \,||\, H^+(pH = 2)|H_2|Pt$
(4) $Pt|H_2|H^+(pH = 4) \,||\, H^+(pOH = 2)|H_2|Pt$

72. Which of the following ion has least ionic mobility in water at 298 K?
(1) Rb^+ (2) K^+
(3) Li^+ (4) Na^+

73. Cell reaction taking place during discharging of lead storage battery is
(1) $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$
(2) $PbO(s) + H_2O_2(aq) \rightarrow PbO_2(s) + H_2O(l)$
(3) $Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$
(4) $PbSO_4(s) + H_2O(l) \rightarrow PbO(s) + H_2SO_4(aq)$

74. Select the correct statement among the following
(1) Molar conductivity of strong electrolyte increases with increase of dilution
(2) Conductivity of weak electrolyte decreases on increase in concentration
(3) Conductivity of strong electrolyte increases with increase of dilution
(4) Molar conductivity increases to two times when cell constant becomes double

75. Select the material of highest conductivity (in S m^{-1}) among the following.
(1) Gold (2) Glass
(3) CuO (4) Ge

76. Example of network solid is
(1) Quartz (2) Iron
(3) NaCl (4) Ar

77. Among the following which crystal system has only primitive as possible variation?
(1) Cubic (2) Tetragonal
(3) Orthorhombic (4) Triclinic

78. If in a cubic unit cell atoms A are present at the corners, atoms B are at the face centres while atoms C are at edge centres, then formula of the compound is
(1) AB_3C (2) AB_3C_3
(3) ABC (4) A_2B_2C

79. If in a close packed system number of atoms are 200 then number of tetrahedral and octahedral voids respectively are
(1) 100 and 200 (2) 200 and 100
(3) 400 and 200 (4) 200 and 400

80. Packing fraction of body centre cubic unit cell is
(1) $\frac{\pi}{2}$ (2) $\frac{\sqrt{3}\pi}{8}$
(3) $\frac{\pi}{8}$ (4) $\frac{\pi}{6}$

81. Number of cation vacancies developed when NaCl is dopped by 10^{-3} mol% $SrCl_2$, is
(1) 6×10^{10} (2) 6×10^{16}
(3) 12×10^{12} (4) 6×10^{18}

82. Select the incorrect match of examples of magnetic property mentioned
(1) Paramagnetic : Fe^{3+}
(2) Diamagnetic : C_6H_6
(3) Ferrimagnetic : $MgFe_2O_4$
(4) Ferromagnetic : MnO

83. Select the incorrect feature of Schottky defect among the following.
(1) It is a vacancy defect
(2) It is shown by ionic substances in which size of cation is much smaller than size of anion
(3) It is observed in high co-ordination number compounds
(4) It is resulted in decrease of density of substance

84. Correct relation of edge length (a) and radius of atom (r) in simple cubic unit cell is
(1) $a = r$ (2) $a = \sqrt{2}r$
(3) $a = \sqrt{3}r$ (4) $a = 2r$

85. AB crystallizes as of NaCl lattice with edge length of 400 pm. The closest distance between two opposite charged ions in the lattice is
(1) 400 pm (2) 100 pm
(3) $100\sqrt{2}$ pm (4) 200 pm

86. Co-ordination number of atoms in ABCABC… type arrangement is
(1) 6 (2) 8
(3) 12 (4) 4

87. Ratio of density of simple cubic and body centred lattice of same element having edge lengths of 20 and 40 pm is
(1) 4 : 1 (2) 1 : 2
(3) 1 : 3 (4) 3 : 4

88. Select the correct statement among the following.
(1) In NaCl lattice, cations are in tetrahedral voids as well as in octahedral voids
(2) In ZnS lattice, anions are in alternate tetrahedral voids
(3) Co-ordination number of carbon in diamond lattice is two
(4) In CsCl lattice, coordination number of Cs^+ ion is 8

89. Number of unit cells present in 2 g molecule of NaBr is (NaBr having similar crystal lattice as of NaCl)
(1) $2N_A$ (2) N_A
(3) $\frac{N_A}{2}$ (4) $\frac{N_A}{4}$

90. If A is occupying all octahedral voids and B is occupying all fcc lattice points, then possible formula of the solid will be
(1) A_2B (2) AB_2
(3) AB (4) AB_3

BOTANY

91. Which of the given phases precedes, event of flowering in life cycle of a plant?
(1) Vegetative phase
(2) Reproductive phase
(3) Senescent phase
(4) Mature phase

92. Read the given statements stating them true (T) or false (F) and select the **correct** option.
(A) Trees continue to increase in height or girth over a period of time
(B) Leaves, flowers and fruits of the same tree not only have limited dimensions but also appear and fall periodically and sometime repeatedly.
(C) All cells of plants are descendents of the zygote
(D) In unfavourable condition, seeds germinate and goes into a period of suspended growth.
(1) A(F), B(T), C(F), D(F)
(2) A(F), B(T), C(T), D(T)
(3) A(T), B(F), C(T), D(F)
(4) A(T), B(T), C(T), D(F)

93. Which of the given statements are **not** true w.r.t. growth in livings?
(A) It is an irreversible process.
(B) It is permanent increase in size of an individual.
(C) Occurs without expense of energy.
(D) It is accompanied by anabolic process, but not catabolic process.
(1) (A) and (B)
(2) (C) and (D)
(3) Only (A)
(4) Only (C)

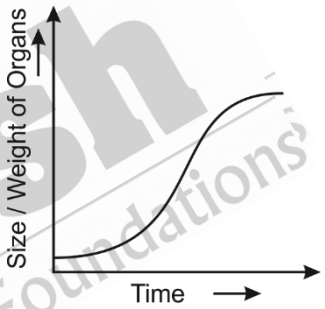
94. The growth of a pollen tube is measured in terms of its
(1) Surface area
(2) Volume
(3) Length
(4) Fresh weight

95. Meristematic cells exhibit all of the given characteristics, **except**
(1) Rich in protoplasm
(2) Abundant plasmodesmatal connections
(3) Undergoes protoplasmic modification and has numerous vacuoles
(4) Possess large conspicuous nuclei

96. Heterophyllous development due to different environment is seen in which of the given plant?
(1) Buttercup (2) Cotton
(3) Coriander (4) Larkspur

97. A gardener is willing to prepare weed-free lawns. Which of the given plant hormone can be effectively employed in such preparations?
(1) Gibberellin (2) Ethylene
(3) Abscisic acid (4) Auxin

98. Select the **incorrect** statement w.r.t. given growth curve.



(1) The curve is a characteristic of cells in culture and many plant organs
(2) It represents typical growth curve in plants
(3) Above curve is obtained for arithmetic growth
(4) It can be mathematically expressed as $W_1 = W_0e^{rt}$

99. Auxin was isolated from tips of coleoptile of oat seedlings by
(1) Francis Darwin (2) Charles Darwin
(3) Skoog and Miller (4) F.W. Went

100. Which of the given plant hormone helps the plants to increase their absorption surface?
(1) Abscisic acid (2) Ethylene
(3) Gibberellins (4) Cytokinin

101. Select the **incorrect** match w.r.t. plant growth regulators (PGRs) and their chemical composition.
(1) Indole compounds – Auxin
(2) Adenine derivatives – Kinetin
(3) Derivatives of carotenoids – Ethylene
(4) Terpenes – Gibberellic acid

102. When the plants are exposed to the necessary inductive photoperiod, the hormonal substance migrates from ___A___ to ___B___ for inducing flowering.
Select the option **correctly** which fills 'A' and 'B'.
(1) A – Leaves, B – Shoot apices
(2) A – Shoot apices, B – Leaves
(3) A – Roots, B – Shoot apices
(4) A – Shoot apices, B – Roots

103. Match column-I with column-II and select the **correct** option

Column-I		Column-II
(A) Auxin	(i)	Increases tolerance of plants to various kinds of stresses
(B) Ethylene	(ii)	Induce parthenocarpy in tomatoes.
(C) Abscisic acid	(iii)	Elongation and improvement of shapes in apples.
(D) Gibberellins	(iv)	Promotes rapid internode/petiole elongation in deep water rice plants

(1) A(ii), B(iv), C(i), D(iii)
(2) A(iii), B(i), C(iv), D(ii)
(3) A(iv), B(iii), C(ii), D(i)
(4) A(ii), B(i), C(iv), D(iii)

104. H.H. Cousins confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas. Later this volatile substance was identified as
(1) Abscissin
(2) Ethylene
(3) Indole-3-acetic acid
(4) Kinetin

105. Which of the given phenomenon refers to promotion of flowering by a period of low temperature?
(1) Vernalisation (2) Photoperiodism
(3) Plasticity (4) Bolting

106. Presence of all of the given chemical substances causes seed dormancy, **except**
(1) Abscisic acid
(2) Phenolic acid
(3) Nitrates
(4) Para-ascorbic acid

107. Cytokinins are **not** abundant in
(1) Root apices
(2) Developing shoot buds
(3) Young fruit
(4) Senescing leaf

108. Which of the given plant hormones promote nutrient mobilisation which helps in the delay of senescence?
(1) Gibberellin (2) Cytokinin
(3) Abscissin (4) Ethylene

109. In most situations, ABA acts as an antagonist to which of the given plant hormones?
(1) Auxin (2) Cytokinin
(3) Gibberellins (4) Ethylene

110. Select the **incorrect** statement w.r.t. life-span.
(1) It is the period from birth to natural death of an organism
(2) Life span of organisms are necessarily correlated with their sizes
(3) A mango tree has much shorter life-span as compared to a peepal tree
(4) It may be as short as a few days or as long as a few thousand years

111. Arrange the given organisms w.r.t. increasing order of their life span and select the **correct** option.
(A) Cow (B) Horse
(C) Parrot (D) Crow
(1) (D) < (A) < (B) < (C)
(2) (A) < (B) < (C) < (D)
(3) (B) < (C) < (A) < (D)
(4) (D) < (C) < (A) < (B)

112. No natural death is seen in all of the given organisms, **except**

(1) *Amoeba* (2) *Penicillium*

(3) *Paramoecium* (4) *Euglena*

113. Genetic variations among offsprings and parents to offsprings are created and inherited during which of the given form of reproduction?

(1) Asexual reproduction

(2) Vegetative reproduction

(3) Binary fission

(4) Sexual reproduction

114. Read the given statements and select the **correct** option.

Statement-A: Reproduction enables continuity of species, generation after generation.

Statement-B: Clones are produced by syngamy.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are correct

(4) Both statements are incorrect

115. Select the **incorrect** match

(1) *Amoeba* – Binary fission

(2) Yeast – Budding

(3) *Chlamydomonas* – Conidia

(4) Sponge – Gemmules

116. During asexual reproduction in which of the given organisms, division of cytoplasm is unequal?

(1) *Paramoecium* (2) Bacteria

(3) *Euglena* (4) Yeast

117. Match the column-I and column-II w.r.t. vegetative propagules in angiosperms.

Column-I	**Column-II**
(A) Ginger	(i) Bulbil
(B) *Agave*	(ii) Offset
(C) *Bryophyllum*	(iii) Rhizome
(D) Water hyacinth	(iv) Leaf buds

(1) A(iii), B(i), C(iv), D(ii)

(2) A(iv), B(ii), C(i), D(iii)

(3) A(iii), B(iv), C(i), D(ii)

(4) A(i), B(ii), C(iii), D(iv)

118. Which of the given statements is **not** true w.r.t. 'terror of Bengal'?

(1) It is an aquatic plant *'Eichhornia*'

(2) Produces cluster of roots above and cluster of leaves below on its horizontal branch

(3) It is one of the most invasive weed found in standing water

(4) It drains oxygen from water body, which leads to death of fishes

119. In banana, vegetative propagation occurs through

(1) Runner (2) Rhizome

(3) Offset (4) Tuber

120. Which of the given plant groups produce isogametes as well as heterogametes?

(1) Algae (2) Bryophytes

(3) Angiosperms (4) Pteridophytes

121. Which of the given plant flowers throughout the life?

(1) China rose (2) Carrot

(3) Rice (4) Wheat

122. Most of the species of *Chara*

(1) Are dioecious

(2) Are protogynous

(3) Have non-jacketed sex organs

(4) Have globule located below nucule on the same plant

123. Which of the given algae produces morphologically similar gametes?

(1) *Volvox* (2) *Ulothrix*

(3) *Fucus* (4) *Chara*

124. In which of the given plant groups, syngamy occurs outside the plant body in external medium?

(1) Algae

(2) Bryophyte

(3) Pteridophyte

(4) Gymnosperm

125. Zoospores produced in algae is

(1) Thick walled zygote

(2) Always diploid

(3) Usually naked

(4) Motile as well as non-motile

126. In which of the given plant groups, embryogenesis is absent?
(1) Algae (2) Bryophyte
(3) Pteridophyte (4) Gymnosperm

127. In flowering plants, fertilized ovule matures into
(1) Seed (2) Fruit
(3) Pericarp (4) Endosperm

128. If number of chromosomes in meiocyte of an individual 'X' is 78. What would be the number of chromosomes in its gametes?
(1) 78 (2) 39
(3) 40 (4) 156

129. In which of the given plants, recovery phase is seen in their life-cycle?
(1) Wheat (2) Rice
(3) Marigold (4) Jackfruit

130. Read the given statements stating them true (T) or false (F) and select the **correct** option.
(A) Members of monera, fungi, algae and bryophytes have haploid plant body.
(B) *Strobilanthes kunthiana* flowers once in 12 years.
(C) *Marchantia* is a monoecious bryophyte
(D) Pollination facilitates transfer of pollen grains to stigma
(1) A(T), B(T), C(F), D(T)
(2) A(T), B(T), C(T), D(T)
(3) A(F), B(T), C(F), D(T)
(4) A(F), B(F), C(T), D(F)

131. Which of the given plant hormone helps to produce chloroplasts in leaves?
(1) Auxin (2) Cytokinin
(3) Gibberellin (4) Ethylene

132. The site of perception of light/dark duration in plants is
(1) Leaf (2) Flower
(3) Stem (4) Lateral bud

133. Auxin was first isolated from
(1) Human urine
(2) Coleoptile tip of canary grass
(3) Coleoptile tips of oat seedlings
(4) Root cap of canary grass

134. Select the **odd** one w.r.t. synthetic auxins
(1) 2,4-D (2) IBA
(3) NAA (4) 2,4,5,-T

135. Which of the given plant hormone decreases sensitivity of roots to gravity?
(1) Auxin (2) GA_3
(3) Ethylene (4) Kinetin

ZOOLOGY

136. Tissue or organ influenced by a hormone is called its
(1) Target (2) Receptor
(3) Gland (4) Messenger

137. Read the following statements carefully and choose the option with all **correct** statements
(a) Hormones are non-nutrient chemicals
(b) Hormones are produced in trace amount
(c) Hormones act as intercellular messengers
(d) Hormones may be proteins, steroids and amines
(1) (a), (b), (c) only (2) (a), (c), (d) only
(3) (b), (c), (d) only (4) (a), (b), (c) and (d)

138. Endocrine cells present in endocrine glands synthesize hormones which are directly secreted into the
(1) Oxygenated blood only
(2) Deoxygenated blood only
(3) Lymph only
(4) Fluid bathing the gland

139. All cells of the body and cellular functions need to be continuously regulated jointly by neural and endocrine system. Although neural system provides a point-to-point rapid coordination among organs, still there is an obvious need of endocrine system, as
(1) Action of neural system is long-lived
(2) Endocrine system is receptor dependent
(3) Nerve fibres do not innervate all cells of the body
(4) Coordination through hormones is faster

140. Which one of the following pairs of organised endocrine organs pour their secretion only in tissue fluids?
(1) Thyroid and parathyroid glands
(2) Pancreas and salivary glands
(3) Heart and Kidney
(4) Gastric glands and Testes

141. Which one of the following is **not** a peptide hormone?
(1) Insulin (2) Oxytocin
(3) Aldosterone (4) Vasopressin

142. Action of thyroxine hormone is similar to action of
(1) Peptide hormones (2) Catecholamines
(3) Androgen (4) ACTH

143. Nuclei are groups of neurosecretory cells in the region of forebrain responsible for secretion of releasing and inhibitory hormones to regulate secretion of
(1) Hypothalamus (2) Adenohypophysis
(3) Adrenal medulla (4) Pineal gland

144. All of the following hormones are derived from amino acids **except**
(1) Melatonin (2) Adrenaline
(3) Norepinephrine (4) Cortisol

145. Select the option which **correctly** matches the endocrine gland with the hormone it secretes and its function

	Endocrine gland	Hormone	Function
(1)	Hypothalamus	Gonadotrophins	Regulates secretion of adenohypophysis
(2)	Pituitary gland	Somatostatin	Inhibits secretion of growth hormone
(3)	Pineal gland	Melatonin	Influences the defence capability of our body
(4)	Thyroid gland	Thyrocalcitonin	Increases blood calcium level

146. Hypothyroidism does **not** causes
(1) Defective development of growing baby
(2) Stunted growth or cretinism
(3) Mental retardation and low intelligence quotient
(4) Protrusion of eyeball and enlarged thyroid

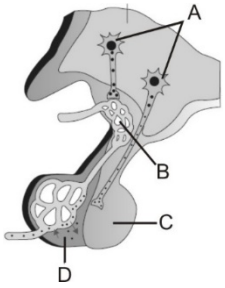
147. Which of the following hormones does **not** influence the process of red blood cell formation?
(1) Erythropoietin (2) Thyroxine
(3) Melatonin (4) Cortisol

148. Secretion of parathormone from parathyroid gland is regulated by
(1) Hypothalamus
(2) Pituitary gland
(3) Autonomous nervous system
(4) Circulating level of Ca^{+2} in blood

149. Which one of the following pairs **correctly** matches a hormone with a disease resulting from its deficiency?
(1) PTH — Tetany
(2) Insulin — Diabetes insipidus
(3) Cortisol — Cushing's syndrome
(4) STH — Acromegaly

150. Which one of the following parts of pituitary gland in adult human is commonly known as anterior pituitary?
(1) Pars distalis (2) Pars nervosa
(3) Pars intermedia (4) Neurohypophysis

151. Observe the diagrammatic representation of pituitary gland and its relationship with hypothalamus given below. Choose the option which represents A, B, C and D **correctly**.



	A	B	C	D
(1)	Hypothalamic neurons	Portal circulation	Posterior pituitary	Anterior pituitary
(2)	Portal circulation	Anterior pituitary	Posterior pituitary	Hypothalamic hormone
(3)	Hypothalamic neurons	Posterior pituitary	Anterior pituitary	Portal circulation
(4)	Anterior pituitary	Posterior pituitary	Hypothalmic neurons	Portal circulation

152. Hormone which causes vasodilation in most of the vital organs, piloerection and glycogenolysis is

(1) Cortisol (2) Thyroxine

(3) Adrenaline (4) Glucagon

153. Read the following statements A and B and select the **correct** answer

A : Insulin hormone is water soluble and directly enters in cell through plasma membrane

B : Oxytocin stimulates contraction of smooth muscle fibres of uterus for parturition

(1) Both statements A and B are correct

(2) Only statements A is incorrect

(3) Only statements B is incorrect

(4) Both statements A and B are incorrect

154. Choose the option which fills the blank **correctly** to complete the analogy w.r.t. glands which synthesises the hormones.

Adrenal cortex : Corticoids :: _____ : Anti-diuretic hormone

(1) Posterior pituitary (2) Pancreas

(3) Hypothalamus (4) Adrenal cortex

155. Given below is an incomplete table about certain hormones, their source glands and one major effect of each on the body in humans. Identify the option which **correctly** fill the blanks A, B and C

Gland	**Secretion**	**Effect on body**
A	Aldosterone	Salt retension
Thyroid gland	**B**	Increase in BMR
Parathyroid gland	PTH	**C**

Options :

	A	**B**	**C**
(1)	Adrenal gland	TCT	Tetany
(2)	Adrenal cortex	Thyroxine	Diuresis
(3)	Testis	TCT	Hypocalcemia
(4)	Adrenal cortex	Thyroxine	Hypercalcemia

156. Asexual reproduction involves

(1) Fusion of male and female gametes

(2) Two parents

(3) Change in the morphology and genetic makeup of progeny

(4) Single parent

157. Encystation is an asexual mode of reproduction in *Amoeba*. It occurs during

(1) Favourable conditions

(2) Unfavourable conditions

(3) In both favourable and unfavourable conditions

(4) Unfavourable condition only after sexual reproduction

158. Choose **odd** one w.r.t ploidy

(1) Germ cell (2) Spermatozoan

(3) Spermatid (4) First polar body

159. The period from birth to natural death of an organism is called

(1) Life style (2) Life expectancy

(3) Maximum life cycle (4) Life span

160. Transverse binary fission in favourable condition occurs in

(1) *Euglena*

(2) *Hydra*

(3) *Amoeba*

(4) *Paramecium*

161. Birds possess cleidoic eggs having a calcareous shell. They have

(1) External fertilization and external development

(2) Internal fertilization and external development

(3) Internal fertilization and internal development

(4) External fertilisation but not development

162. Select **correct** match w.r.t. animals and their approximate life-span

(1)	Butterfly	—	1-2 months
(2)	Crow	—	5 years
(3)	Parrot	—	140 years
(4)	Crocodile	—	90 years

163. Select the **incorrect** statement w.r.t. reproduction
(1) Reproduction enables the continuity of species generation after generation
(2) Asexual reproduction may occur with or without involvement of gamete formation
(3) Clones are morphological and genetically similar individuals
(4) In monerans and protists, cell division is not a mode of reproduction

164. Gametogenesis and gamete transfer are
(1) Pre-fertilization events
(2) Fertilization events
(3) Post fertilization events
(4) Events during senescence

165. A haploid parent produces gamete by
(1) Mitosis
(2) Meiosis
(3) Amitosis
(4) Both mitosis and meiosis

166. Chromosome number in meiocyte of butterfly is 380. What will be chromosome number in gametes of butterfly?
(1) 180 (2) 190
(3) 90 (4) 270

167. Read the following statements carefully and choose the option with all **correct** statements.
(a) Fusion of gametes is most vital event of sexual reproduction
(b) In parthenogenesis, male gamete undergoes development to form new organism without fertilization
(c) In internal fertilization, syngamy occurs in the body of organism
(d) Zygote formed may be diploid or haploid in case of parthenogenesis
(1) (a) and (b) (2) (b) and (c)
(3) (c) and (d) (4) (a) and (c)

168. Which of the following is considered as the vital link that ensures continuity of organisms of one generation and the next?
(1) Zygote (2) Sperm
(3) Ova (4) Embryo

169. Which of the following is responsible for transitions between the three phases of life in both plants and animals?
(1) Lipids
(2) Carbohydrates
(3) Hormones
(4) Enzymes

170. Adenohypophyseal hormone, ACTH mainly controls the secretion of
(1) Mineralocorticoids
(2) Glucocorticoids
(3) Entire adrenal cortex
(4) Entire adrenal gland

171. Which of the following is **not** function of Androgens?
(1) Erythropoiesis
(2) Maturation and function of male accessory sex organs
(3) Development of mammary glands in males
(4) Male sexual behaviour

172. Pituitary gland secretes how many hormones?
(1) 6 (2) 7
(3) 8 (4) 9

173. Read the following statements A and B and choose the **correct** answer.

Statement-A : Hormones from thyroid gland play an important role in development and maturation of central neural system.

Statement-B : Thymosins play a major role in differentiation of T-lymphocytes and production of antibodies.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

174. Which of the following is **not** the function of glucocorticoids?
(1) Lipolysis
(2) Gluconeogenesis
(3) Proteolysis
(4) Glycogenolysis

175. Match column-A with column-B and choose the option with all **correct** match.

	Column-A		**Column-B**
(a)	Atrial natriuretic factor	(i)	Maintains pregnancy
(b)	Cholecystokinin	(ii)	Regulates sexual behaviour in females
(c)	Estrogen	(iii)	Decrease blood pressure
(d)	Progesterone	(iv)	Increases secretion of pancreatic digestive enzymes

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

176. Which of the following hormones act by the interaction of hormone-receptor complex with the genome?

(1) Aldosterone (2) Adrenaline

(3) GH (4) TSH

177. Choose **odd** one w.r.t. disease caused due to hyposecretion of hormones

(1) Diabetes insipidus

(2) Simple goitre

(3) Addison's disease

(4) Acromegaly

178. During puberty, all of the following are functions of androgenic steroids secreted by Zona reticularis **except**

(1) Growth of axial hairs

(2) Growth of pubic hairs

(3) Growth of facial hairs

(4) Formation of primary sex organs

179. Which of the following is **not** a function of LH?

(1) Induces ovulation of Graafian follicle

(2) Maintains corpus luteum

(3) Induces leydig cells to secrete androgens

(4) Acts on cells of Sertoli for spermatogenesis

180. Immune responses of old persons become weak due to degeneration of

(1) Pineal gland

(2) Thymus

(3) Corpus luteum

(4) Gonads

❑ ❑ ❑

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